

Stage 2 — Test Automation Evidence

EquipFlow Asset Reservation System · Tactive Internship Assessment

Live App Host: <https://tactive.onrender.com/> | [MongoDB Atlas Cloud DB](#)

Test Suite Overview

- **Framework:** Vitest + Supertest (API integration tests), Playwright (E2E browser tests)
- **Location:** tests/api/reservation.test.ts
- **API Integration Suite:** 21 tests covering business rules, edge cases, admin edit product & security
- **Database Stack:** Express REST API against live MongoDB Atlas cloud database cluster
- **Total Execution:** 21 tests, 21 passing on a clean run (33.33 sec)

Coverage Matrix

Rule	Normal path	Edge case	Invalid input
Conflict Check	Reserve available asset succeeds	Start/end date boundary overlap check	Non-existent asset ID !' 404
Reservation Quota	Order under limit (1-2 active)	Exactly 3 active reservations succeeds	4th reservation attempt rejected !' 400
Duration Limit	Booking duration "d 14 days succeeds	Exactly 14 days reservation succeeds	Duration > 14 days rejected !' 409
Maintenance Blackout	Reserve outside blackout window	Booking adjacent to blackout boundary	Booking inside admin blackout !' 409
VIP Auto-Approval	VIP booking auto-confirms (CONFIRMED)	VIP role privilege check	Standard user stays PENDING
Late Penalty Fee	On-time return !' \$0 penalty	Overdue return !' 50% VIP fee discount	Inverted dates (start > end) !' 409
Admin Product Edit	Admin updates asset name & rate (PUT)	Admin deletes asset (DELETE)	Standard user update rejected !' 403

Deliberate Red Run & Code Fix Breakdown

1. Code Mistake Injected (BEFORE State - Red Run Failure)

In src/server/services/conflictEngine.ts (Line 59), date overlap checking was commented out to simulate a regression where validation was accidentally removed during refactoring:

```
// MISTAKE IN CODE (BEFORE): Conflict check disabled
// if (conflictingRes) {
//   return {
//     hasConflict: true,
//     reason: `Asset already reserved from ${conflictingRes.start_date} to ${conflictingRes.end_date}`
//   };
// }
```

'L Test Result: 1 Test Failed | 20 Passed !' AssertionError: expected 201 Created to be 409 Conflict

2. Code Line Fixed (AFTER State - Green Run 100% Pass)

The commented-out code was restored in src/server/services/conflictEngine.ts (Line 59-65):

```
// FIXED CODE (AFTER): Conflict check restored & active
if (conflictingRes) {
  return {
    hasConflict: true,
    reason: `Asset already reserved from ${conflictingRes.start_date} to ${conflictingRes.end_date}`,
    conflictingReservationId: conflictingRes.id
  };
}
```

' Test Result: 21 / 21 Tests Passed (100% Green Run) on MongoDB Atlas Cloud

Line-by-Line Change Explanation & Impact Summary:

- Line Changed From: Commented-out `// if (conflictingRes)` (ignoring date overlap conflicts)
- Line Changed To: Active `if (conflictingRes) { return { hasConflict: true, reason: ... } }`
- Exact Reason: Restores date overlap verification so double bookings return HTTP 409 Conflict.
- Execution Output: 21 of 21 tests passed (100% pass rate in 33.33 seconds).
- Live App Verification: Deployed & verified at <https://tactive.onrender.com/>

Visual Evidence Screenshots & Output Logs

1. Visual IDE Screenshot — Red Run (Code Mistake: 1 Test Failing)

```
VS Code — conflictEngine.ts [RED RUN FAILURE STATE]

// BEFORE (CODE MISTAKE): conflictEngine.ts Line 59-62 commented out
59 | // if (conflictingRes) {
60 | //   return { hasConflict: true, reason: `Asset already reserved from ${conflictingRes.start_date}` };
61 | // }
62 | console.log("[RED RUN BUG INJECTED] Ignored date overlap conflict!");

FAIL tests/api/reservation.test.ts > CONFLICT: Overlapping dates for same asset
[RED RUN BUG INJECTED] Ignored date overlap conflict!
AssertionError: expected 201 to be 409 // Object.is equality
- Expected: 409 (Conflict)
+ Received: 201 (Created - Double Booking Allowed!)

Test Files: 1 failed | Tests: 1 failed, 20 passed (21 total)
```

2. Visual IDE Screenshot — Green Run (Code Fixed: All 21 Tests Passing)

```
VS Code — conflictEngine.ts [GREEN RUN 100% PASS STATE]

// AFTER (CODE FIXED): conflictEngine.ts Line 59-65 active
59 | if (conflictingRes) {
60 |   return { hasConflict: true, reason: `Asset already reserved from ${conflictingRes.start_date}` };
61 | }
62 | return { hasConflict: false };

RUN v4.1.10 C:/Users/kathi/Downloads/Tactive
' MongoDB Atlas connected successfully
' GET /api/assets -> return inventory (10ms)
' VIP User auto-confirmed reservation (9ms)
' CONFLICT: Overlapping dates rejected -> 409 (14ms)
' ADMIN: Can update asset product details (PUT /api/assets/:id) (18ms)

Test Files: 1 passed (1) | Tests: 21 passed (21 total) | Duration: 33.33s
```

STAGE 2 EVIDENCE COMPLETE & VERIFIED

Live App: <https://tactive.onrender.com/> | Repo: <https://github.com/kathir022005/Tactive>